

ECE436 Final Project Proposal:

IoT Connected SmartHub Dashboard

Thiago Costa, Johannes Hirt, and Anar Amartuvshin

Introduction

The purpose of this project is to design and implement an IoT based smart dashboard using multiple ESP32 devices. The system displays real time data such as room temperature, humidity, and ambient light level, while also providing user interaction with a display, hosting a web server for remote interactions and being able to control external devices.

Overall, the design involves three ESP32 devices using low power wireless technologies, LoRa for long distance communication and Bluetooth Low Energy (BLE) for short distance communication. We are also utilizing other communication protocols like I2C or SPI for communication between internal modules. Sensor data is collected from each ESP32 device to the central SmartHub or our main ESP32 device. Also, users can monitor data and control devices via the screen.

In addition, the system will enable users to remotely control other devices, such as a lamp. As an extended feature we have included gesture control through a camera. The display will be able to rotate using a servo.

Overall this IoT project involves wireless communication, embedded control, user interface, and interacting with multiple sensors.

Objectives and Specifications

- Use three ESP32 devices each for screen, main controller, and camera.
- Enable wireless communication using both BLE and one LoRa technologies for short and long distance communications.
- Provide User Friendly UI for monitoring and control.
- The screen should turn off when the user is not present.
- Must report data such as temperature, humidity, and light level from sensors.
- Main ESP32 LoRa node powered by a simple battery. (I think we need a 5V battery, the battery we have in the classroom is 3.7 A).
- Based on hand gestures the display rotates 90 degrees using a servo.
- Include a website based interface for remote control.
- Extra: If we have time, we can include user authentication and different applications

Project Strategy

This project will start with simple setup for each small component and testing simple communication using LoRa and BLE, and small hardware setup for proper functionality and

avoiding complicated debugging. Once individual devices work, we will test the combined whole system for the project.

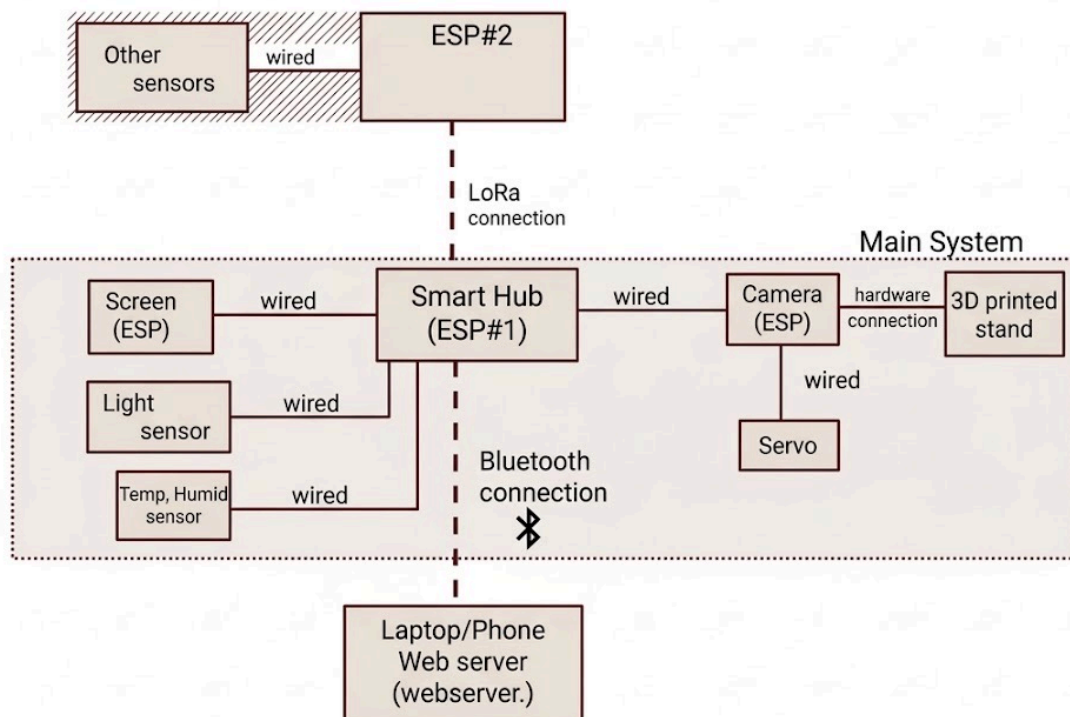


Figure 1: System design final draft.

We start with a small setup for all devices. We are planning to set up the camera first, and testing with simple sensor interaction and display. Then we can also set up the simple LoRa communication between the outside sensor (local temperature sensor) and main controller (Smart Hub). Also, in parallel, set up for BLE communication between camera, display screen and main ESP controller. For testing purposes, we will test if we read correctly by reading the data from sensors and ensuring its functionalities.

For powering the ESP32 devices, we are planning to use a simple battery or usb charger. So far, we are planning to use 3 ESP32 devices as noted in the figure1 above.

For mechanical design, we are using a servo that can hold about 5 kilograms, and we will program it to rotate +/-90 degrees based on camera data. Thus, we will need a screen stand and we are planning to simply 3D print it. and attach it to the servo and screen.

In the end, we integrate all subsystems, and run testing, and cleaning the codes.

plan:

Sensors:

- Camera (Motion Detection)
- Proximity (Detecting if someone is gets near the dashboard to turn screen on)
- Photoresistor (adjust screen brightness or turn it off)

- Humidity (Simply recording data about the room it is in)
- Temperature (Same as humidity)
- Microphone (probably not too realistic)

Actuators/Other components:

- Screen (Leading into a 7+ inch display)
- Servo Motor (To turn the screen into landscape or portrait mode)

Interface:

- Website that will be able to display data and control the dashboard and/or the devices connected to it (Work almost like a phone app, but on a website)
- Website will also be responsible for sending phone notifications +
- BLE + LoRa communication with external devices
- Ability to send commands to devices or perform other actions

User Interface:

- Home screen that would look like a phone screen (Wallpaper, time, date, etc) and would have a low brightness mode if we could change screen brightness
- UI should change between Portrait and Landscape modes
- User will be able to access different apps by swiping to the side
- Weather Forecast app (pulling data from OpenWeatherAPI)
- Calendar App (connected to an API for syncing) with events + reminders
- Alarm functionality
- Scores for World Cup games!!! (OPTIONAL)
- Snake Game with gestures (OPTIONAL)

Gestures:

- Swipe (Left, Right, Up and Down)
- Rotate screen (should be a toggle between portrait and landscape for simplicity)

Task and Priorities

[Task 0] Design choices and ordering components

Possible Components (with links):

- Screen: [7 inch 800x480 ESP32 screen](#)
- Camera: [Xiao ESP32 S3 Camera](#)
- Motor: [20Kg RC Servo Motor](#)
- Photoresistor: Probably the one we have
- Environment Sensor (Pressure, Temperature and Humidity at once): BHT11

[Task 1] Easy set up and check if all components are functional

- Screen set up
- Camera set up
- Humidity sensor set up
- Temperature sensor set up
- Servo motor set up
- (UI) Decide what screen should show?

- (UI) How does the user interact with it?
- (UI) How is data displayed and updated?
- Simple BLE connection showing bluetooth work.
- LoRA: simple transmission and receiver communication check between two ESP32.
- Calibration, error calculation, and setting thresholds for sensors and cameras.

[Task 2] BLE and sensor assembly and communication

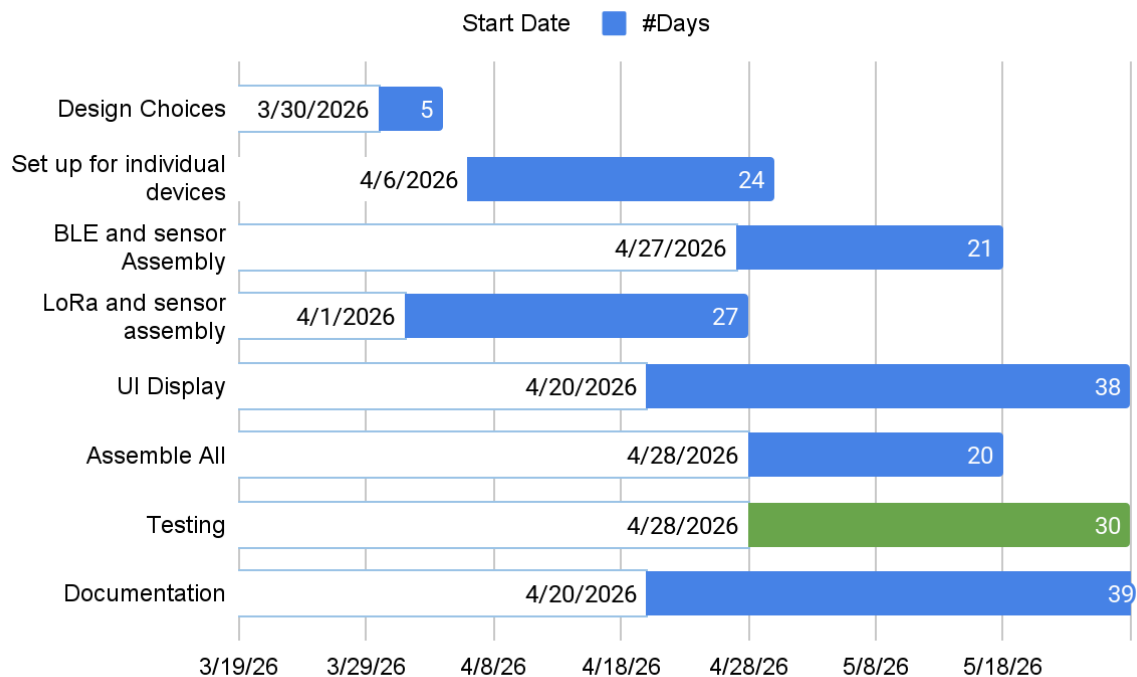
[Task 3] LoRa and sensor Assembly and communication

[Task 4] UI display

[Task 5] Assemble All

[Task 6] Testing

[Task 7] Documentation



Research Plans

- We need to look into how to setup the screen and render the UI with LVGL or a similar library
- We need to look into how to rotate the UI between landscape and portrait mode, since LVGL projects have a set resolution
- We need to look into how to setup the ESP Camera, run existing models on it and how to develop our own in order to detect the gestures and send commands
- We need to look into how to design a 3D printed enclosure that will ensure that the servo is able to rotate the screen without it falling
- Look into how we can establish LoRa and Bluetooth communication between different devices
- We need to look into how to setup the webserver and allow it to remotely control the dashboard and other external devices connected to it
- How to access the APIs that we need

Note: Please modify the link and copy paste from there .Thank you ! For BOM and scheduling.You should have an access to it and able to edit!

<https://docs.google.com/spreadsheets/d/1plel2BHulZoLRZGKzDnG3d2uCqxQxBP8o8vX621AbMU/edit?usp=sharing>

Schedule

Budget

Item	Details	Costs (\$)
Screen:	7 inch TFT Screen for displaying the UI	53.49
Camera	ESP32 based camera module for gesture recognition	21.83
Motor	Servo motor used to rotate the screen	14
Photoresistor	Record data about the room's lighting	
Humidity Sensor	Measure humidity	
Temperature Sensor		
total		89.32
Cost per person		29.77333333